

AI-Based Glue Route Quality Inspection for Industrial Metallic Parts - Guidelines

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Introduction

This guide explains how to set up, run, and maintain the glue-inspection system. The solution uses a YOLOv11 model for defect detection (Good vs NG) and a Flask web app with three pages: Upload Image, Live Detection, and Capture & Save. It supports local development (CPU/GPU) and edge deployment on NVIDIA Jetson Orin Nano (TensorRT optimized).

Folder & Files Structure

```
glue_inspection_app/
├── app.py          # Main Flask application
├── weights/
│   └── yolov11.pt  # Trained YOLO model weights
├── templates/      # HTML pages
│   ├── base.html  # main codes for the design
│   ├── index.html # Home page
│   ├── upload.html # Upload Image page
│   ├── live.html  # Live Detection page
│   └── capture.html # Capture & Save page
├── static/         # CSS/JS/Images
│   ├── styles.css/ # Styling files
│   └── app.js/      # JavaScript files
│   ├── hero.svg/   # hero image in the homepage
│   └── logo.svg/    # System Logo image
├── uploads/        # User-uploaded images
├── results/        # Output with detection results
└── new_images/     # Captured images for retraining
```

1. app.py – Runs the Flask web application and links the YOLO model with the interface.

2. yolov11.pt – The trained deep learning model (weights).
3. templates/ – Contains HTML files for the front-end pages.
4. static/ – Contains CSS and JavaScript for styling and interactivity.
5. uploads/ – Stores uploaded images from the Upload page.
6. results/ – Stores detection results (annotated images).
7. new_images/ – Stores captured images for dataset expansion.

Steps to Run the Application

1- Open the terminal

2- Enter the project folder

```
cd ~/ glue_inspection_app
```

3- Create a virtual environment

```
python3 -m venv .venv  
source .venv/bin/activate
```

4- Install dependencies

```
pip install flask ultralytics opencv-python torch torchvision
```

5- Run the Flask app

```
python app.py
```

6- Open in browser

Troubleshooting

Problem: Flask app doesn't start

- Check if Python installed and also check the version (≥ 3.8).
- Install all requirements

Problem: No detections / always blank

- Ensure yolov11.pt is in the project root.
- Lower confidence threshold inside app.py (e.g., `conf=0.25`).

Problem: Camera not working

- For Jetson CSI camera, make sure to use GStreamer pipeline and if it is correct in app.py.
- Reset the camera service

```
sudo systemctl restart nvargus-daemon
```

Problem: Port already in use

- Run this command in the terminal

```
pkill -f "python app.py" 2>/dev/null || true
```